

Low-Cost Geometric Directions in Frozen Speech Models: Where a Training-Free Deepfake-Hardness Signal Helps—and Where It Does Not

Anonymous authors

Paper under double-blind review

Abstract

Deepfake-speech detectors are inconsistent in two systematic ways: against a fixed detector some synthesis systems evade detection far more often than others, and a detector trained on one recording domain often fails on another. Working entirely at small scale—one GPU, a frozen $\sim 100\text{M}$ -parameter encoder, and no detector retraining—we show that much of this inconsistency is legible in the *geometry* of the frozen representation. Inside the frozen encoder there is a single natural-versus-synthetic direction, and how far a synthesis system’s utterances *spread* along it predicts how hard that system is to detect—the only quantity, out of 83 we tested, to survive a global false-discovery-rate correction. Two further results follow. First, the direction is not universal: it rotates across datasets to near-orthogonality (and reverses sign), the geometric reason detectors fail to transfer. Second, it is a training-free corrective signal whose value is conditional on headroom: it reduces a weak detector’s error by $\approx 21\%$ while yielding no measurable change for an already-strong one. Every quantitative claim is independently audited, with confirmed and exploratory results separated. The contribution is a precise account of where low-cost geometric directions in frozen models help—and where they do not.

1 Introduction

Synthetic speech is now realistic enough that its detection is a security necessity, and the standard defence is a classifier that flags synthetic utterances. Yet detectors are unreliable in two ways that aggregate accuracy hides: against a fixed detector some synthesis systems evade far more often than others, and a detector tuned on one recording domain often fails on another. Neither had a clean explanation.

We provide one, and we obtain it at small scale. Modern detectors sit on top of a large frozen self-supervised (SSL) speech encoder (Chen et al., 2022); we treat that frozen representation as the object of study, before any decision boundary is drawn. Our central finding is a **natural-versus-synthetic axis**: a single direction running from natural to synthetic speech. A system whose utterances cluster tightly along this axis is detected easily; one whose utterances spread along it is hard. That spread, sd_{along} , is our predictor.

Three results follow, each established at small scale. (i) *The law*: sd_{along} predicts per-system hardness and is the single quantity, among 83 tested, to survive global multiple-testing correction (Sec. 3). (ii) *The axis rotates*: each dataset has its own axis pointing a different way, so a transferred axis is uninformative—the geometric reason transfer fails (Sec. 4). (iii) *A training-free corrective signal, conditioned on headroom*: fusing the axis into a detector’s score requires no retraining and no labels and improves detection, but only when the detector has headroom (Sec. 5). We hold a high evidential bar throughout: an independent audit recomputed every number and is reported in-line (Sec. 6).

2 Setup

Detector. Our primary detector (WavLM-GAT) is the phoneme-level graph-attention model of Zhang et al. (2025): a frozen WavLM encoder feeds adaptive phoneme pooling, a 3-layer 6-head graph-attention network (Velićković et al., 2018) with skip connections, a BiLSTM, and a linear head (Fig. 1). The encoder is frozen ($\sim 95\text{M}$ parameters); only the $\sim 13\text{M}$ -parameter head trains. For architecture generality we also use AASIST (Jung et al., 2022).

Corpora and hardness. The hardness law is established on MLAAD (Müller et al., 2024) (61 text-to-speech systems with ≥ 8 utterances). The training-free lever and the axis-rotation check additionally use ASVspoof 2019 LA (Wang et al., 2020). Per-system hardness is $1 - \text{AUC}$ of the detector’s scores for that system against a shared bona-fide pool; on a held-out check the metric has split-half reliability 0.76 (R^2 ceiling ≈ 0.86).

Axis. Per corpus we estimate a natural-versus-synthetic direction w in mean-pooled frozen WavLM-L12 space as the (unsupervised) bona-fide—synthetic centroid difference, leave-one-system-out so a held-out system never informs its own axis. For each system we record sd_{along} , the standard deviation of its per-utterance projections onto w .

Compute. Everything runs on a single GPU; the largest single training run is $\approx 10^{16}$ FLOPs (well under a 10^{20} budget), and all geometry re-runs from cached embeddings in minutes.

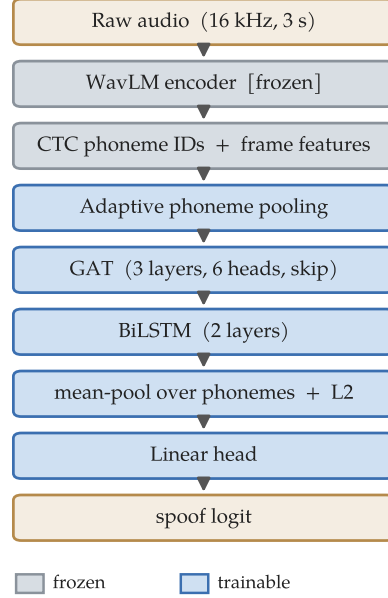


Figure 1: The WavLM-GAT detector, redrawn; architecture after Zhang et al. (2025). A frozen WavLM encoder yields phoneme IDs and features; a small trainable head (adaptive phoneme pooling, GAT, BiLSTM, linear) emits a spoof logit. All our geometry is measured in the frozen layer-12 space.

3 A Natural–Synthetic Axis Predicts Detection Hardness

Across 61 MLAAD systems, per-system hardness rises with sd_{along} (Fig. 2a; Spearman $\rho = 0.60$, $p < 10^{-6}$; leave-one-system-out $R^2 = 0.277$, permutation $p = 2.5 \times 10^{-4}$; bootstrap ρ CI [0.41, 0.73]). A system whose utterances straddle the boundary—some appearing essentially genuine to the encoder—is difficult to detect, and those near-genuine utterances are the ones that evade.

This is the most robust predictor we identified. Of 83 quantities tested across the study, sd_{along} is the *only* one to survive a global Benjamini–Hochberg correction. It survives leave-one-system jackknifing (ρ range 0.58–0.63) and strengthens as measurement noise falls: $\rho = 0.60 \rightarrow 0.64 \rightarrow 0.69$ at $\geq 8/12/16$ utterances per system—the attenuation signature of a real effect. Against the 0.86 reliability ceiling it captures $\approx 32\%$ of the *explainable* hardness variance: a partial but robust account.

Directionality, not circularity. Because a centroid axis can resemble a detector’s own logit direction, we verified that the mechanism is directional, not an artifact of magnitude. Substituting the top discriminative directions of the representation with class-blind values degrades detection ($\Delta\text{AUC} = -0.032$), whereas the identical operation on the equal-magnitude orthogonal residual does not ($+0.004$); the contrast is consistent across all three seeds (we do not claim significance at $n = 3$). An earlier ungated version of this intervention inflated a spurious “compactness” effect by $\approx 80\%$; closing that leak isolates the directional mechanism. Finally, the rank-order law replicates on a different architecture: on AASIST fine-tuned on MLAAD, sd_{along} still ranks hardness ($\rho = +0.35$, $p = 0.006$). We are

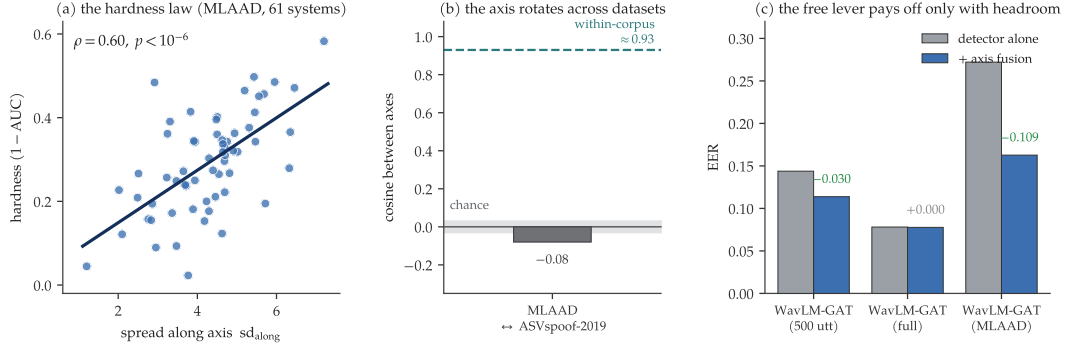


Figure 2: **(a)** Spread along the corpus-internal natural–synthetic axis predicts MLAAD hardness across 61 systems. **(b)** The axis rotates: the MLAAD and ASVspoof-2019 internal axes are near-orthogonal and slightly negative ($\cos \approx -0.08$), far below within-corpus reliability (≈ 0.93 , dashed) and inside the high-dimensional chance band (≈ 0.03 , grey). **(c)** Training-free axis fusion reduces a weak WavLM-GAT’s error ($\Delta\text{EER} -0.030$, and -0.11 for an under-trained MLAAD detector) but yields no change for an already-strong one ($+0.000$).

precise that only the rank-order form transfers—the R^2 regression form does not on AASIST ($r^2 \approx 0$)—and report both.

4 The Axis Rotates Across Datasets

Why does a predictor (or detector) fit on one corpus fail on another? Because the direction it relies on is not the same direction elsewhere. Each corpus’s internal axis is estimated almost noiselessly (split-half cosine ≈ 0.93), yet the axis learned on MLAAD and the axis learned on ASVspoof-2019 are near-orthogonal—and in fact slightly negative ($\cos \approx -0.08$; Fig. 2b), against a 768-dimensional chance floor of ≈ 0.03 . A transferred axis therefore points, if anything, the wrong way. The operational lesson is direct: an axis-based hardness diagnostic must be re-estimated on the target recording domain, and a detector’s per-system error profile from one corpus should not be trusted on another. We cannot fully exclude that part of the rotation reflects recording-channel nuisance rather than synthesis semantics; either way the lesson is unchanged.

5 Training-Free Axis Fusion Helps Only Under Detector Headroom

Because the axis identifies *which* systems a detector will struggle with, its projection can be fused into the detector’s decision, $z(\text{logit}) + \lambda z(\text{axis})$, with no retraining and no new labels. Whether this helps depends entirely on headroom (Fig. 2c). We trained a deliberately weak WavLM-GAT on only 500 utterances; the unsupervised axis reduced its equal-error rate from 0.144 to 0.114 ($\Delta\text{EER} = -0.030$, 95% CI $[-0.046, -0.013]$, $p \approx 0$), improving all five attack-disjoint folds—a $\approx 21\%$ relative reduction. The *identical* procedure applied to a strong, near-ceiling WavLM-GAT (EER 0.078) produced no change distinguishable from noise ($\Delta\text{EER} = -0.0004$ to $+0.004$, $p \approx 0.56$ – 0.80). In domain on MLAAD, where the base detector is likewise far from ceiling, the same zero-training centroid fusion improves EER from 0.272 to 0.163 (axis alone 0.187), negative across all folds and seeds. Axis fusion therefore provides a practical, training-free improvement precisely when the base detector is undertrained, and none when it is already strong; in both cases the deployed detector is left untouched.

6 Independent Audit of the Reported Claims

Every quantitative claim was checked by an independent audit that recomputed each number from cached artifacts, returning a tally of 8 verified, 6 mislabeled, 5 misleading, and 2 contradicted claims; each was corrected or removed. The corrections are substantive, not cosmetic: a cross-corpus cosine reported as $+0.36$ was in fact negative (the axis rotates with the *wrong* sign); an R^2 of 0.316 attributed to sd_{along} was in fact a three-feature triplet model, the correct single-feature value being 0.277; a “small residual causal effect” disagreed in sign across seeds and was dropped; and “13 interventions” was corrected to the 22 conditions actually run. We report the audit in-line and separate confirmed from exploratory claims throughout.

7 Limitations and Conclusion

The law is a text-to-speech phenomenon: on voice-conversion attacks all predictors give $\rho \approx 0$, consistent with converted speech inheriting its content from source speech. All geometry uses a single SSL backbone (WavLM); generality to wav2vec 2.0 or HuBERT is untested. The law rests on $n = 61$ systems and the causal contrasts on only three seeds, so the directional mechanism is trustworthy in sign but not powered as a significance claim. Cross-dataset *prospective* prediction of hardness—using one corpus’s axis to rank a never-seen corpus’s systems in advance—remains unconfirmed and is left as future work. Within these bounds, the contribution is a reusable finding for small-scale research: a low-cost, frozen, training-free geometric direction is a genuine tool for understanding and partially correcting deepfake detectors, and we have delimited precisely where it applies (weak detectors, within a dataset) and where it does not (across datasets, or on already-strong detectors).

Reproducibility statement

All results come from a single-GPU pipeline with a frozen $\sim 95\text{M}$ -parameter encoder and a $\sim 13\text{M}$ -parameter trainable head ($\ll 3\text{B}$ parameters); the largest single training run is $\approx 10^{16}$ FLOPs. We report estimated FLOP budgets and release logs, checkpoints, cached embeddings, and the full audit, with scripts that regenerate every reported number.

References

- Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, et al. WavLM: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1505–1518, 2022.
- Jee-weon Jung, Hee-Soo Heo, Hemlata Tak, Hye-jin Shim, Joon Son Chung, Bong-Jin Lee, Ha-Jin Yu, and Nicholas Evans. AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 6367–6371, 2022.
- Nicolas M Müller, Piotr Kawa, Wei Herng Choong, Edresson Casanova, Eren Gölge, Thorsten Müller, Piotr Syga, Philip Sperl, and Konstantin Böttinger. MLAAD: The multi-language audio anti-spoofing dataset. In *International Joint Conference on Neural Networks (IJCNN)*, pp. 1–7, 2024.
- Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations (ICLR)*, 2018.
- Xin Wang, Junichi Yamagishi, Massimiliano Todisco, Héctor Delgado, Andreas Nautsch, Nicholas Evans, Md Sahidullah, et al. ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech. *Computer Speech & Language*, 64:101114, 2020.

- 167 Kuiyuan Zhang, Zhongyun Hua, Rushi Lan, Yushu Zhang, and Yifang Guo. Phoneme-level
168 feature discrepancies: A key to detecting sophisticated speech deepfakes. In *Proceedings of*
169 *the AAAI Conference on Artificial Intelligence*, volume 39, pp. 1066–1074, 2025.